

JEE MAIN REVISION MASTER: LIMITS, CONTINUITY AND DIFFERENTIABILITY

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 29 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Limits: L'Hopital's Rule & Standard Trigonometric Limits	15 PYQs	Limits
High	Continuity & Differentiability at a Point	14 PYQs	Continuity & Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Limits: L'Hopital's Rule & Standard Trigonometric Limits	$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$, $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$, $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$
Continuity & Differentiability at a Point	$\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = f(c)$, $\text{Differentiability} \implies \text{Continuity}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Limits: L'Hopital's Rule & Standard Trigonometric Limits (Very High Yield — 15 PYQs)

Core Principle: Evaluation of limits in indeterminate forms ($0/0$, ∞/∞) using L'Hôpital's Rule (differentiate numerator and denominator separately). Standard limits: $\lim_{x \rightarrow 0} (\sin x / x) = 1$, $\lim_{x \rightarrow 0} (\tan x / x) = 1$, $\lim_{x \rightarrow 0} (1 - \cos x / x^2) = (1/2)$, $\lim_{x \rightarrow 0} (1+x)^{1/x} = e$, $\lim_{x \rightarrow 0} (a^x - 1 / x) = \ln a$.

■ **Caution / JEE Pitfall:** L'Hôpital's Rule applies ONLY to indeterminate forms $0/0$ and ∞/∞ ; check form before differentiating!

■ **Speed Trick:** In KCET, for $(1)^\infty$ forms: $\lim_{x \rightarrow a} [f(x)]^{g(x)} = e^{\lim_{x \rightarrow a} g(x)[f(x) - 1]}$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "If $f(x) = x^2 - 1$ if $x \geq 2$ and $x + 1$ if $x < 2$, then $\lim_{x \rightarrow 2} f(x) =$..." [Ans: B]
- [2026 Shift] Q#2 (MCQ): "If $\lim_{x \rightarrow 0} (\frac{x^2 - ax - 3bx - 3}{x}) = 5$, then $a + b =$..." [Ans: C]
- [2026 Shift] Q#3 (MCQ): "If $f(x) = \begin{cases} ax + 7 & \text{if } x < 1 \\ 3x - 1 & \text{if } x = 1 \\ b/x + 3 & \text{if } x > 1 \end{cases}$ is continuous at $x = 1$..." [Ans: D]

3.2 Continuity & Differentiability at a Point (High Yield — 14 PYQs)

Core Principle: A function $f(x)$ is continuous at $x=c$ if $\text{LHL} = \text{RHL} = f(c)$. Differentiability: $f'(c)$ exists if $\text{LHD} = \text{RHD}$.

Fundamental theorem: Every differentiable function is continuous, but the converse is NOT true (e.g. $f(x) = |x|$ is continuous at $x=0$ but not differentiable due to sharp corner). Greatest integer function $[x]$ is discontinuous at all integers.

■ **Caution / JEE Pitfall:** Finding constant k so $f(x)$ is continuous: simply equate the expressions of the two branches at the junction point and solve for k !

■ **Speed Trick:** Polynomial, exponential, sine, and cosine functions are continuous and differentiable everywhere on $\mathbb{R}$.

Representative Source PYQs:

- **[2025 Shift] Q#1 (MCQ):** "Match the following: In the following, $[x]$ denotes the greatest integer less than or equal to x . Column - I Column - II (a) $x | x | x | x \dots$ " **[Ans: N/A]**
- **[2025 Shift] Q#2 (MCQ):** "The function $f(x) = e^x + ax, x < 0$ $b(x-1)^2, x \geq 0$ is differentiable at $x=0$. Then..." **[Ans: N/A]**
- **[2025 Shift] Q#3 (MCQ):** "\$\text{A function } f(x) = \begin{aligned} &\frac{e^x}{x} - 1 \\ &e^{\frac{1}{x}} + 1 \end{aligned}\$, & \$\text{if } x \neq 0\$ & \$\text{if } x = 0\$ end{aligned}..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

If $f(x) = x^2 - 1$ if $x \geq 2$ and $x + 1$ if $x < 2$, then $\lim_{x \rightarrow 1} f(x) + \lim_{x \rightarrow 2} f(x) =$

- (A) 3
- (B) 5 **[Correct]**
- (C) 7
- (D) 9

Correct Answer: B

Solution Analysis: For $x \rightarrow 1$, the values of x are near 1, which is less than 2. So we use $f(x)=x+1$. Hence, $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (x+1) = 1+1=2$. Now for $x \rightarrow 2$, we check both sides. For $x < 2$, $f(x)=x+1$. So, $\lim_{x \rightarrow 2^-} f(x) = 2+1=3$. For $x \geq 2$, $f(x)=x^2-1$. So, $\lim_{x \rightarrow 2^+} f(x) = 2^2-1=4-1=3$. Since both left-hand and right-hand limits are equal, $\lim_{x \rightarrow 2} f(x) = 3$. Now, $\lim_{x \rightarrow 1} f(x) + \lim_{x \rightarrow 2} f(x) = 2+3=5$. So the correct

Benchmark Problem 2: KCET 2025 (Q#1)

Match the following: In the following, $[x]$ denotes the greatest integer less than or equal to x . Column - I Column - II
 (a) $|x| \times |x|$
 (b) $|x| + |x| + \sqrt{|x|}$
 (c) $x + [x]x + [x]x + [x]$
 (d) $|x-1| + |x+1|$

(i) continuous in $(-1, 1)$
 (ii) differentiable in $(-1, 1)$
 (iii) strictly increasing in $(-1, 1)$
 (iv) not differentiable at, at least one point in $(-1, 1)$

- (A) a-i, b-ii, c-iv, d-iii
- (B) a-iv, b-iii, c-i, d-ii
- (C) a - ii, b - iv, c - iii, d - i
- (D) a - iii, b - ii, c - iv, d - i

Correct Answer: None

Solution Analysis: (a) $x|x|$ $f(x) = x^2$, $x \geq 0$ $-x^2$ $x < 0$ differentiable in $(-1, 1)$ (b) $\sqrt{|x|} = \sqrt{x}$, $x \geq 0$ $\sqrt{-x}$, $x < 0$ Not differentiable at $x = 0$ (c) $x + [x]$ strictly increasing in $(-1, 1)$ (d) $|x-1| + |x+1| = \begin{cases} -x+1-x-1 & x < -1 \\ -x+1+x+1 & -1 \leq x \leq 1 \\ 2x & x > 1 \end{cases}$ Continuous $(-1, 1) = \blacksquare \blacksquare \blacksquare \blacksquare \blacksquare \blacksquare$ $-2x$, $x < -1$ $2x$, $x > 1$